

# Raghav Senthil Kumar

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## EDUCATION

### University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

## EXPERIENCE

### Founding Software Engineer | Clusion

September 2025 – January 2026

- Shipped working software in an *Agile* delivery team of **10 engineers**, joining code reviews while building the *Python* (*FastAPI*) search engine for an *AWS*-hosted high school debate research platform with **10K+ site visitors**
- Designed *REST APIs* that index **600K+ documents** on *Elasticsearch* and return results in **under 1 second**
- Wrote a text-retrieval algorithm from scratch that extracts evidence segments from debate cases by validating each source URL and filtering statistical outliers with **IQR** on the word counts of the tag, citation and evidence text
- Refactored the search index by de-duplicating it, removing the **30%** of documents that were identical copies

### Product Development Intern | Jetson

January 2025 – February 2025

- Consulted directly with the CEO, delivering a **5-page report** with strategies to optimize user engagement
- Ran **7 rounds of UI/UX testing** on 3 experimental features for **100K+ users**, applying feedback each round
- Redesigned onboarding in *Figma* from **12 pages to 5** and validated the change via **A/B testing** with 30 users

### Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **200+ TikTok posts** from 3 years in *Python* (*pandas*), *Power BI* and *Excel* to find outreach patterns
- Discovered “science fun facts” posts earned 69.3% of all views despite being just 7% of what we published
- Delivered business value with a strategy the Executive Director adopted, lifting engagement **24% (8K+ views)**

## PROJECTS

### Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Wrote clean, well-tested *Python* and *JavaScript* code for a full-stack scheduling app (*React*, *Firebase*) that auto-books meetings for groups of **30+ people** on Google Calendar, owning the product from design to delivery
- Designed a **5-step scheduling algorithm** that scans availability, ranks time slots and emails confirmations
- Kept the code well-tested with **14 unit and integration tests** covering every step of the booking workflow
- Integrated the Google Calendar API and built a lobby that authenticates invitees and queues follow-up meetings
- Licensed the app to **8 startup clients** with **2 LOIs (\$3.5K)**. **Winner, 2025 Congressional App Challenge**

### SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built *Python* (*FastAPI*) *REST APIs* that ingest live data from 3 edge AI tools into **4 Random Forest models**
- Delivered actionable steps through a **Gemma Voice LLM** on a *Next.js* dashboard, a modern development stack
- Led a team of 5 from ideation through delivery and final presentation. **USAII Global Hackathon Finalist**

### Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Designed the data model for paper records in a relational *PostgreSQL* database hosted on *Supabase*, then wrote *SQL* queries that let users filter the stored papers by their research field, publication year and total citation count
- Built integrations with 2 *REST APIs* (*OpenAlex*, *Semantic Scholar*) to ingest and backfill paper metadata
- Researched emerging AI tools to deploy a *Claude Sonnet 4.5* **RAG** agent over session-scoped vector documents

### Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Headed a **4-person** client build of Kamion's (**YC S22**) pricing tool. **2nd Place at UIUC Founder's Hackathon**
- Debugged the agent's *JSON schema* after every misread until **10 images in a row** were correct on 3 features
- Forecast truck prices from the extracted attributes using *quantile regression*, scoring **87%** accuracy across 30 samples

## SKILLS

**Programming Languages:** Python, Java, JavaScript, SQL, C++, HTML/CSS

**Frameworks/Platforms:** AWS, Google Cloud, PostgreSQL, MySQL, FastAPI, React, Node.js, Next.js, Firebase

**DevTools:** Git, GitHub, Docker, CI/CD, VS Code, Claude Code, Codex, Figma, Bun, Arduino IDE, Kaggle

**Concepts/Practices:** Object-Oriented Programming, REST APIs, API Development, Unit Testing, Agile, ETL